

Jack White

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Education

Purdue University

West Lafayette, IN

B.S. Computer Science and B.S. Artificial Intelligence

2023–2027

GPA: 3.6/4.0 • Dean's List (6 times)

Coursework: Algorithms, Operating Systems, Machine Learning, Robotics, Imitation Learning (current)

Experience & Research

BPM Microsystems

Houston, TX

Automation / AI Software Engineering Intern

Jul–Aug 2025; May–Aug 2026

- Built **AlgoBuilder**, an **end-to-end agentic AI system powered by large language models (LLMs)**, to generate and validate NAND/NOR flash programming algorithms; saved an estimated **10–15 engineering hours per algorithm** with **over 80% fully automated completion** across **more than 40 validated algorithms**.
- Modernized **Chip**, an internal AI assistant used by **approximately 75% of employees (40–50 people)**; integrated Salesforce/Google Drive through Model Context Protocol (MCP), added tests, and reduced model-call costs.
- Built **PDR Builder**, an AI-assisted pipeline that turns datasheets and product specs into engineering artifacts with deterministic validation and human review; passed **27/27 historical Tier-C design validations**.
- Upgraded **BPWin REST API 1.2–1.9** with job observability, device discovery, remote dialog handling, NAND support, and validation endpoints for unattended automation.
- Delivered multiple presentations on **Chip, AlgoBuilder, and PDR Builder** usage and workflows.

Purdue University – Prof. Yexiang Xue

West Lafayette, IN

Robot Learning Research – Zero-Shot & Active Learning

Present

- Developing LLM and diffusion-policy methods to generate demonstrations without human input, bridging hub-to-hub and hub-to-goal transitions in latent space for zero-shot robot learning.

Purdue University – Prof. Yung-Hsiang Lu

West Lafayette, IN

Computer Vision Research

Aug–Dec 2025

- Implemented computer-vision methods for piano-to-MIDI transcription on **Rach3 and PianoYT** to evaluate generative music-performance systems.

Selected Projects

ControVirtual – Robot Learning and Teleoperation

Apr 2026

- Fine-tuned **SmolVLA**, a vision-language-action model, on **50 teleoperated episodes**, raising task success from **0% to approximately 75%**. Integrated Meta Quest/Unity, OpenCV, and a digital twin; earned **Honorable Mention – Best Use of AMD Tech, StarkHacks 2026**.

SlopMop – AI-Content Detection Browser Extension

2026

- Implemented **LinkedIn, Facebook, Instagram, X, and Reddit** integrations for a browser extension that flags likely AI-generated content; reduced automatic detection time to **approximately 0.6 seconds**.

TradingSim + RCBot – Market Simulation and Automated Trading

2026

- Built an event-driven simulator and autonomous strategy engine with Level-2 order-book data, latency- and depth-aware fills, risk controls, backtesting, and Interactive Brokers paper trading.

Technical & Professional Skills

Programming: Python, C, C++, C#, Java, TypeScript/JavaScript, SQL, Assembly, data structures

AI / ML: PyTorch, Hugging Face, LightGBM, LLM APIs, pandas, NumPy, OpenCV

Web / Backend: React, Redux, Vue, Angular, FastAPI, Flask, REST APIs, OAuth, WebSockets

Cloud / Data: AWS, cloud computing, Docker, PostgreSQL, Redis, NoSQL

Engineering / Tools: full-stack software engineering, Linux, Git, Mercurial, Datadog, Unity, GitHub Copilot

Professional: data visualization, UX patterns, product specs, communication skills, technical writing, customer value

Additional Experience

Professional Keyboardist – Down to Funk, 3rd Street Collective, Silver Wizard. Opened for **Magdalena Bay** with **3rd Street Collective** at Purdue's Elliott Hall of Music for approximately **6,000 attendees** (Mar 2026).